MemeSense: A Social-Intuition Augmentation Interface for Blind and Low-Vision Users in Mobile Messaging

1. Design Overview

Mobile messaging increasingly depends on emojis, reaction images, and short, context-dependent phrasing to convey tone (e.g., teasing, sarcasm, warmth). Blind and low-vision (BLV) users can access text via screen readers or refreshable braille displays but still face a gap: many social cues are *implicit* and are not conveyed well by literal readouts or basic image descriptions. This is amplified by the fact that image descriptions are often missing in user-generated content; for example, a large-scale study of Twitter images found descriptions were present only in a tiny fraction of image tweets [1][2].

In this context, an augmentation interface is the closed-loop interaction layer that (i) senses messaging content, (ii) maps it to a small set of actionable social-cue labels with uncertainty handling, and (iii) delivers those cues through low-bandwidth feedback (braille/audio/haptics) under user control. MemeSense augments *social intuition* by summarising tone and intent rather than only reading literal content.

2. Why this idea

Current accessibility workflows often provide *literal* information (e.g., object-centric image descriptions) but do not reliably deliver social intents such as irony or teasing. Memes illustrate this problem clearly: they frequently combine template-specific visual context with short text, and research has shown meme accessibility is challenging precisely because conventional alt text is hard to author at scale and often absent [3 - 5].

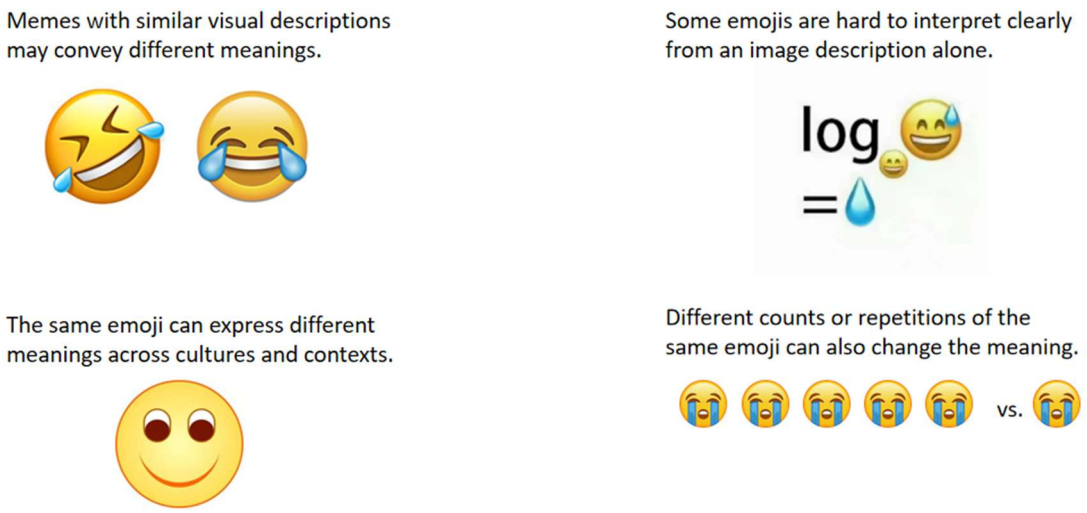

Figure 1: Examples showing why literal image/emoji descriptions can miss meaning: similar visuals may imply different intent, meanings vary by culture and context, some emojis are ambiguous, and repetition changes interpretation.

In practice, BLV users also commonly rely on separate assistive apps (e.g., image recognition software, camera-based describers or "ask an AI" tools) to interpret images. This creates friction because the user must switch context or invoke extra steps. For example, Seeing AI and Lookout are standalone camera/recognition apps designed to describe the visual world [6][7]. VoiceOver image descriptions require enabling specific settings and using "Explore Image" actions, which is helpful but still interaction-heavy and not tailored to meme/social intent [8]. Be My AI similarly involves capturing/sharing an image and then asking follow-up questions in a separate interaction flow [9].

Therefore, the design goal is not to replace screen readers, but to add a compact layer that helps users rapidly infer tone/intent during real chats.

3. Augmented Function and Interface Design

3.1 What it augments

MemeSense augments three cues that are high value in chat and feasible to constrain:

1. Opinion or stance: positive, neutral, negative
2. Pragmatic mode: literal vs joking/teasing vs sarcastic/ironic
3. Intensity: low, moderate, high. Implying calm or escalated.

This scope is intentionally narrow to avoid overclaiming and to make evaluation measurable. Emoji accessibility research shows that emoji descriptors and platform differences can hinder

communication for BLV users, motivating a lightweight “meaning layer” rather than raw emoji readouts [5][10].

3.2 Interface Design

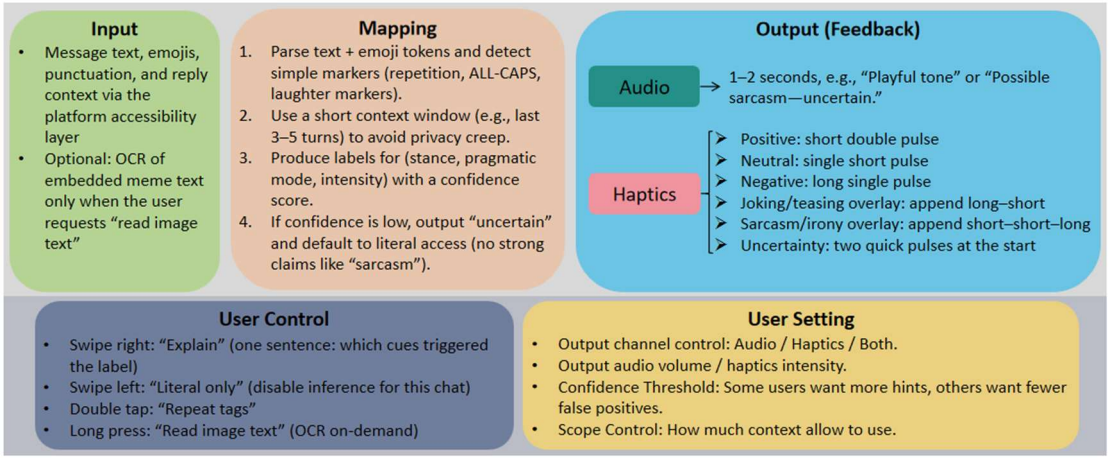

Figure 2: MemeSense interface design pipeline.

Figure 2 summarises the MemeSense augmentation interface as a closed loop from input to feedback. The system first captures message text, emojis, punctuation, and reply context via the platform accessibility layer, while running OCR on embedded meme text only when the user requests it. It then maps these signals into a small set of social-cue labels (stance, pragmatic mode, and intensity) using a short context window to reduce privacy creep. Each output includes a confidence score; low-confidence cases are explicitly marked as “uncertain” and default to literal access rather than strong claims (e.g., sarcasm). Feedback is delivered through 1–2 second audio cues and a learnable haptic codebook. Users can quickly request an explanation, repeat the tags, switch to literal-only mode, and adjust channels, sensitivity, and context scope.

4. Development Plan and Evaluation

4.1 Development Plan

The development plan includes following four crucial stages:

- Prototype UI flow and output formats (audio phrases, haptic patterns)
- Implement content capture via Accessibility APIs; add OCR as an optional tool
- Train/validate the cue classifier on messaging-style data; implement confidence gating
- Integrate user controls (literal-only toggle, explain-on-demand) and privacy settings

4.2 User Evaluation

A usability study with 8–12 blind/low-vision participants, comparing:

a) Baseline screen reader only.
b) Screen reader + MemeSense.

Tasks: interpret short chat snippets (including emojis and sarcasm), decide intended tone, and choose an appropriate reply option. Metrics:

- Tone/intent accuracy
- Time to reach a decision
- Self-reported confidence and workload
- Error severity (e.g., misreading sarcasm as hostility)

5. Risks and Ethics

Misclassification can cause social harm (for example, wrongly marking a friendly joke as hostile). MemeSense reduces this risk by using confidence thresholds, clear "uncertain" signals, and an easy switch to literal-only mode. Privacy risk is reduced by limiting the context window and offering on-device processing; any cloud use should require clear consent and should not store message content by default.

6. Conclusion

MemeSense extends conventional accessibility beyond literal access by augmenting a small, high-impact subset of social cues (valence, pragmatic mode, intensity). Through a controlled interface loop—sensing, bounded inference, and low-bandwidth braille/audio/haptic feedback—it aims to improve social comprehension and confidence for blind and low-vision users in everyday messaging.

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